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Paper Code: TCE 813

B.Tech (Civil Engineering)
End Semester Examination 2025
VIII Semester
Hydropower Engineering

Time: Three Hours

MM: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions in each main question.
- (iii) Total marks for each main question is **Twenty**

Q1

(10X 2 = 20 Marks)

- a) A hydropower station has a rated capacity of 90 MW and works with a variable river flow rate throughout the year. If the river discharge reduces by 40% during dry seasons (4 months per year), find:
 - a) The new power output in dry months.
 - b) The average power generated throughout the year.
 - c) The Load Factor of the plant.
- b) A runoff river station with an installation capacity of 15000 Kw operated at 15% load factor, when it serves at peak load station. What would be the lowest discharge in the stream so, that the station may serve as a base load station. It is given that the plant efficiency is 75% when working under head of 20m. Also calculate the maximum load factor of the plant when the discharge in the stream rises to $20\text{m}^3/\text{s}$. CO1
- c) Discuss the relative advantages and disadvantages of the hydropower and the thermal power plant?

Q2

(10X 2 = 20 Marks)

- a) Calculate the specific speed of a turbine and suggest the type of turbine required for a river having a discharge of 240L/s with an available head of 45m. Take the efficiency of turbine = 82% and speed = 450r.p.m
- b) Explain the Diversion Canal Plant and its different layouts with a neat sketch? CO2
- c) In a hydroelectric station, water is available at the rate of $175\text{m}^3/\text{s}$ under a head of 18m the turbine run at speed of 150 r.p.m with overall efficiency of 82%. Find the number of turbine required if they have the maximum specific speed of 460.

Q3

(10X 2 = 20 Marks)

- a) Describe all the three main divisions of a power station with a neat sketch
- b) Describe underground power station? Explain the history of underground power station? CO3
- c) Describe the ways of locating an underground power station with neat sketch?

Q4

(10X 2 = 20 Marks)

- a) A penstock, with an internal diameter of 1.20 m, supplies water at a head equivalent to 17.6 kg/cm^2 . There is a possibility of 20 per cent increase in the pressure due to transient conditions. The design stress and the efficiency of the joint may be assumed to be 1020 kg/cm^2 and 85 percent respectively. Calculate the approximate wall thickness of the penstock required. CO4
- b) A power canal, bed width 15 m, may be assumed to be rectangular in shape with a steady state depth of flow of 3 m. The canal supplies water to a power house with 3 turbines,

each turbine rated at a discharge of $30\text{m}^3/\text{s}$. If the load in the power house is suddenly thrown off so that two of the turbines have to be shut down, what would be the height of the surge in the canal?

- c) Derive the expressions for the water hammer pressure in case of a rigid pipe and in case of an elastic pipe.

Q5

(10X 2 = 20 Marks)

- a) A Francis turbine on a major hydro-power project in Pakistan is to operate under a net head of 150 m the desired output per turbine is 175 MW and the synchronous generator speed is 136.4 rpm. Model studies indicate an overall efficiency of 91.5 percent and the prototype efficiency is likely to be at least 0.5 percent. Hyd. Efficiency is estimated as 96 percent. Calculate the specific speed and the discharge to the machine. Also suggest suitable values of exist angle of guide vanes, the vane angles of the wheel , the inlet and outlet diameter of the wheel and the breadth of the wheel at the inlet. Make suitable assumptions.

CO5

- b) Describe the way you distinguish different turbines on the basis of hydraulic features?
- c) Explain a draft tube and how is draft tube applicable in the installation of turbines?